INVIQTVS Submission Report

EPFL x Quandela Quantum Challenge 2026

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Abstract

This report documents the final submission pipeline developed by Team INVIQTVS for the EPFL x Quandela quantum challenge on swaption surface modeling. The implemented system addresses two operational tasks defined by the challenge workflow: (i) six-step forward forecasting of volatility surfaces and (ii) imputation of two rows with partially missing entries in the test template. The codebase is organized as a reproducible pipeline with four executable stages, a judge-facing Streamlit application, and versioned artifacts for traceability.

Methodologically, we benchmark classical forecasters (Ridge, XGBoost, LSTM, and Naive Persistence), quantum-inspired and photonic-hybrid approaches (QRC, QKGP, QRLSTM), and a reconstruction branch based on quantum and classical autoencoders. The official benchmark relies on challenge data only and uses time-ordered validation; an additional random-walk (RW) experiment is included only as an auxiliary sanity check and is not used for leaderboard claims.

Empirically, the strongest official one-step baseline in our current artifacts is Naive Persistence (MAE 0.002556), while the most competitive quantum forecaster is QRLSTM (MAE 0.002721). In reconstruction, the quantum autoencoder outperforms the matched classical bottleneck. The report concludes with a technical discussion of why this ranking is coherent with literature on short-horizon persistence, kernel methods, reservoir computing, and variational photonic models, and provides concrete next steps for production-grade extensions.

1 Introduction

Forecasting and reconstructing swaption volatility surfaces is a high-dimensional time-series problem with direct relevance for derivatives pricing and risk management. In market practice, robust baselines are typically hard to beat for very short horizons, yet non-linear and quantum-inspired methods may provide gains in representation quality, uncertainty handling, and robustness under distribution shifts.

Our submission was designed under three constraints:

1. **Faithfulness to the official workflow**: benchmark conclusions are drawn only from the

challenge dataset.

2. **Reproducibility**: a one-command pipeline (`python run_pipeline.py`) regenerates all training artifacts and the final `results.xlsx`.

3. **Comparability**: classical and quantum families are evaluated with consistent preprocessing and metrics.

From a modeling perspective, the project combines ideas from term-structure modeling and modern machine learning. We reduce surface dimensionality through PCA [1], use regularized linear and sequence baselines [2–4], and benchmark photonic quantum layers and kernels motivated by current QML literature [5–8]. For financial context, we treat persistence and random-walk behavior as strong null models [9–11].

2 Problem Definition and Data

2.1 Challenge Tasks

The submission must fill a provided test template with:

- six future volatility surfaces (forecast horizon),
- two dates with partially missing entries (imputation).

2.2 Dataset Structure

The training matrix has shape $\mathbf{494} \times \mathbf{224}$: 494 dates and 224 grid points per surface. Each surface point corresponds to a $(\text{tenor}, \text{maturity})$ pair in a fixed 14×16 grid. The loader normalizes schema and parses metadata from canonical challenge headers.

2.3 Official vs Auxiliary Validation

Two evaluation regimes are maintained by design:

1. **Official workflow**: all model selection and ranking for submission rely only on challenge data.

2. **Auxiliary RW check**: a separate script compares model forecasts to synthetic random-walk future surfaces; this is explicitly labeled as non-official.

3 Pipeline and Experimental Protocol

3.1 End-to-End Pipeline

The repository executes four deterministic stages:

1. `02_classical_baselines.py`
2. `03_quantum_reservoir.py`
3. `04_quantum_autoencoder.py`
4. `05_final_predictions.py`

Artifacts are stored in `trained_models/`, and the final workbook is written to `data/results.xlsx`.

3.2 Preprocessing

Given raw prices $x_{t,j}$, we apply z-score normalization per grid point:

$$z_{t,j} = \frac{x_{t,j} - \mu_j}{\sigma_j}. \tag{1}$$

Then PCA is fitted on training-only normalized surfaces:

$$\mathbf{z}_t \approx \mathbf{W}_k \mathbf{p}_t, \quad k = 6 \tag{2}$$

for forecasting branches. Windowed samples use lookback $W = 20$, horizon $H = 1$, and temporal split ratio 85%/15%.

3.3 Metrics

We track MAE, RMSE, MAPE, R^2, and maximum absolute error:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{3}$$

Forecast and reconstruction are reported separately to avoid conflating tasks.

4 Classical Forecasting Baselines

4.1 Ridge Regression

Ridge is trained on flattened PCA windows. For target vector $\mathbf{y}$ and design matrix X, parameters solve:

$$\hat{\beta} = \arg\min_{\beta} \|\mathbf{y} - X\beta\|_2^2 + \lambda\|\beta\|_2^2. \tag{4}$$

This baseline is strong for low-noise short-horizon settings [2].

4.2 XGBoost

We train one gradient-boosted regressor per output PCA coordinate. XGBoost provides flexible non-linear fits with strong tabular performance [4], though at this horizon it does not outperform persistence.

4.3 LSTM

An LSTM sequence model consumes 20×6 PCA windows and predicts the next 6-D PCA state. LSTMs remain a standard recurrent baseline for temporal dependencies [3].

4.4 Naive Persistence

The naive predictor copies the last observed PCA state forward. In financial time series this is a critical benchmark and often strong at one-step horizon [9].

5 Quantum and Hybrid Approaches

5.1 QRC: Photonic Reservoir + Ridge Readout

QRC uses a fixed photonic circuit as non-linear feature map (MerLin/Perceval backend), then trains a classical Ridge readout. We run an ablation over $(n_{\text{modes}}, n_{\text{photons}}, \text{encoding})$. Best official setting in artifacts is:

$$n_{\text{modes}} = 8, \quad n_{\text{photons}} = 3, \quad \text{encoding} = \texttt{last}.$$

Reservoir memory capacity is measured over lags 1–30; total memory in the saved run is 1.0745, indicating non-trivial temporal retention.

5.2 QKGP: Quantum Kernel Gaussian Process

QKGP uses a photonic feature map to build kernels and then applies GP regression per PCA coordinate:

$$\hat{\mu}_* = K(X_*, X)\,[K(X, X) + \alpha I]^{-1}\,\mathbf{y}, \tag{5}$$

$$\hat{\sigma}_*^2 = K(X_*, X_*) - K(X_*, X)\,[K(X, X) + \alpha I]^{-1}\,K(X, X_*). \tag{6}$$

This model is motivated by quantum-kernel literature [5–7] and classical GP forecasting theory [12].

5.3 QRLSTM: Quantum-Inspired Reservoir Features + LSTM

QRLSTM builds fixed quantum-inspired reservoir states and feeds their temporal trajectories to an LSTM head. In our artifacts it is the strongest quantum forecaster (official one-step MAE), with near-classical baseline accuracy.

5.4 QAE and Classical AE

The autoencoder branch models reconstruction and latent temporal dynamics:

- **Quantum AE**: classical encoder → photonic bottleneck → classical decoder.
- **Classical AE**: same macro-architecture with classical bottleneck for controlled ablation.

This setup follows the quantum autoencoder paradigm [13] and standard representation learning principles [14].

6 Imputation and Final Submission Output

Missing-row imputation combines local temporal neighbors with masked reconstruction (QAE branch when available). Future rows are generated from forecast artifacts. The pipeline stage producing `results.xlsx` currently assembles forecasts through an ensemble object for workbook generation, while the judge-facing Streamlit view prioritizes best-model interpretability in dedicated pages.

7 Empirical Results

7.1 Official One-Step Forecast Metrics: Classical

Table 1: Classical baselines on official one-step validation (original price space).

Model	MAE	RMSE	MAPE (%)	R^2	Max Error
Naive Persistence	**0.002556**	**0.003477**	**1.3796**	**0.998733**	**0.018030**
Ridge	0.002847	0.003880	1.5974	0.998423	0.018050
LSTM	0.004169	0.005914	2.4928	0.996336	0.032804
XGBoost	0.004202	0.006085	2.3614	0.996121	0.030313

7.2 Official One-Step Forecast Metrics: Quantum/Hybrid

Table 2: Quantum and hybrid forecasting models on official one-step validation.

Model	MAE	RMSE	R^2	Notes
QRLSTM	**0.002721**	**0.003637**	**0.998538**	hybrid reservoir + LSTM
QRC	0.015715	0.019664	0.959495	best ablation: 8 modes, 3 photons, last
QKGP	0.021049	0.027211	0.922440	fidelity kernel GP

7.3 Reconstruction Branch (Separate Task)

Table 3: Autoencoder reconstruction metrics (validation, original price space).

Model	MAE	RMSE	R^2	Trainable Params
Quantum AE	**0.003556**	**0.004727**	**0.997600**	31,370
Classical AE	0.004028	0.005700	0.996511	31,390

Latent temporal forecasting for the QAE branch achieves MAE = 0.016621 and $R^2 = 0.964869$ in latent space.

7.4 Auxiliary RW Validation (Non-Official)

Table 4: Auxiliary validation against synthetic RW future surfaces (not used for official ranking).

Model	MAE	RMSE	R^2
Naive Persistence	**0.000265**	**0.000325**	**0.999988**
Ridge	0.000877	0.001194	0.999843
QRLSTM	0.002264	0.003328	0.998776
QAE	0.003253	0.004101	0.998142
QRC	0.027342	0.029825	0.901727
QKGP	0.034276	0.040472	0.819038

8 Discussion Against State of the Art

8.1 Why Naive Can Beat Sophisticated Models at One Step

At short horizons, surface dynamics can be highly persistent. This makes one-step forecasting close to a local random-walk regime, where persistence is difficult to beat consistently. Our results align with this literature-based expectation in financial forecasting settings [9, 10].

8.2 Interpretation of Quantum Model Ranking

QRLSTM is competitive because it combines expressive non-linear reservoir features with sequence modeling capacity. QKGP is theoretically attractive due to uncertainty quantification [12], but practical GP complexity and sample caps can limit performance. QRC remains interpretable and useful for ablation, but in this dataset the fixed readout stack is not the top forecaster.

8.3 Reconstruction vs Forecasting

The QAE branch is valuable for representation learning and imputation quality, not necessarily for winning one-step forecasting. Keeping this separation explicit avoids metric misinterpretation and yields a cleaner technical narrative.

8.4 Hardware Realism and Noise Considerations

The photonic layer choices are aligned with near-term variational photonic computation frameworks and known limitations (trainability, noise, barren plateaus) [8, 15]. Our archived experiments include hooks for noisy simulations and additional circuit variants, supporting future hardware-aware studies.

9 Complexity and Scalability Analysis

9.1 Asymptotic View

From an engineering perspective, the current stack spans linear, tree-based, recurrent, and kernelized learners:

- Ridge readouts are efficient and stable for high-dimensional linearized windows.
- XGBoost complexity scales with tree count and depth, usually practical at this dataset size.
- LSTM/QRLSTM training scales with sequence length and hidden size and remains tractable here.
- QKGP is the least scalable in sample count due kernel matrix operations.

This explains part of the observed trade-offs: methods with richer uncertainty structure (GP) can become sample-constrained, while low-variance short-horizon tasks strongly reward simple and stable predictors.

9.2 Practical Runtime Considerations in This Repository

The implementation already includes practical controls:

1. capped GP samples in `fit_from_pca(max_samples=220)`,
2. deterministic artifact caching (`*.npy`, `*.pkl`, `*.json`),
3. separate training phases to avoid recomputation in the judge app.

Table 5: Qualitative complexity profile of implemented model families.

Family	Main training cost driver	Inference profile	Scalability note
Ridge / Naive	linear algebra on flattened windows	very fast	strong baseline for short horizon
XGBoost	number of trees, depth, features	fast to moderate	robust tabular nonlinearity
LSTM / QRLSTM	epochs $\times$ sequence length $\times$ hidden size	moderate	good temporal expressiveness
QRC + Ridge	quantum feature extraction + linear readout	moderate	feature cost dominates for large batches
QKGP	kernel matrix build/inversion	moderate to high	sample-limited due GP scaling
QAE branch	autoencoder epochs + latent temporal fit	moderate	useful for reconstruction/imputation

10 State-of-the-Art Mapping

Table 6 makes explicit how each implemented component connects to established literature.

Table 6: Implemented components and corresponding state-of-the-art references.

Component	Role in this submission	Main references
PCA compression	low-dimensional latent factors for forecasting windows	[1]
Ridge baseline	regularized linear predictor in PCA/window space	[2]
XGBoost baseline	non-linear tree ensemble baseline	[4]
LSTM baseline	recurrent temporal baseline in PCA space	[3]
QRC	fixed photonic reservoir + classical readout	[16, 17]
QKGP	quantum-kernel feature map + GP regression	[5–7, 12]
QRLSTM	hybrid quantum-inspired reservoir features + sequence model	[3, 17]
QAE / Classical AE	reconstruction and latent dynamics branch	[13, 14]
Temporal validation	time-ordered splitting and forecasting evaluation	[18]
Financial baseline interpretation	persistence / random-walk consistency checks	[9–11]

11 Engineering, Reproducibility, and Submission Readiness

11.1 Reproducibility Checklist

- Clean entrypoint: `python run_pipeline.py`
- Deterministic artifact locations: `trained_models/`, `data/results.xlsx`
- Streamlit judge view: `streamlit run app.py`
- Cross-platform setup in repository docs (Windows and Linux/macOS commands)

11.2 Repository Hygiene

Unused/legacy resources were moved to `archive/` and documented. The active submission path is kept concise and traceable for judges.

12 Limitations and Next Steps

1. **Hyperparameter breadth**: expand search for PCA dimension, lookback, and model-specific regularization.
2. **Uncertainty reporting**: calibrate predictive intervals for QKGP and hybrid models, and assess reliability diagrams.
3. **True multi-step training**: include direct multi-horizon training for all model families, not only selected branches.
4. **Hardware-aware validation**: extend noisy photonic simulations and robustness stress tests.
5. **Scalability**: benchmark latency and memory under increased sample counts and model ensembles.

13 Conclusion

The submission delivers a complete and reproducible benchmark of classical and quantum approaches for swaption surface forecasting and imputation. On the official one-step workflow, Naive Persistence remains the best classical baseline, while QRLSTM is the strongest quantum/hybrid forecaster. The quantum autoencoder improves reconstruction over a matched classical bottleneck, supporting the value of quantum-enhanced latent representations for surface reconstruction tasks.

Overall, the main contribution is not a single universally dominant model, but a rigorous comparative framework: transparent preprocessing, consistent metrics, clear separation between official and auxiliary validation, and a judge-facing interface that maps code artifacts to interpretable outcomes.

A Reproduction Commands

```
python -m venv .venv
# Windows
.venv\Scripts\Activate.ps1
# Linux/macOS
source .venv/bin/activate
pip install -r requirements.txt
python run_pipeline.py
streamlit run app.py
```

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